THE ZEROS OF ζ AND OF DIRICHLET L-FUNCTIONS IN THE FLUCTUATIONS OF PRIME PAIRS AND PRIME TRIPLETS: A NUMERICAL STUDY

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ABSTRACT. Let $W_{\mathcal{H}}(x) = \sum \prod_i \log(p + h_i)$ count, with logarithmic weights, the primes $p \le x$ for which every $p + h_i$ is prime, for an admissible pattern $\mathcal{H} = (h_1, \dots, h_k)$. The Hardy–Littlewood conjecture predicts $W_{\mathcal{H}}(x) \sim \mathfrak{S}(\mathcal{H})\, x$. A first-order heuristic, in which each member of the pattern carries the local fluctuation of the primes in its residue class, predicts that the oscillating part of $W_{\mathcal{H}}(x) - \mathfrak{S}(\mathcal{H})x$ is $\mathfrak{S}(\mathcal{H}) \sum_\chi c_\chi(\mathcal{H})\, \psi(x, \chi)$, with explicit coefficients $c_\chi(\mathcal{H})$ given by averages of characters over admissible residue classes. In particular the zeros of ζ enter with weight $k\, \mathfrak{S}(\mathcal{H})$, and the zeros of $L(s, \chi)$ enter with a weight whose sign depends on $\mathcal{H}$. We test this prediction numerically, with no fitted parameter, using an exact sieve up to 10^{10} (and up to 10^{12} for $h = 2, 4, 6$), and a matched filter that projects the fluctuations onto the explicit templates $(\psi(x) - x)/\sqrt{x}$ and $\psi(x, \chi)/\sqrt{x}$ in the frequency band $50 \le \gamma \le 6000$. For the thirty gaps $h = 2, \dots, 60$, the ζ term is found at 0.98 ± 0.04 times its predicted size, and twin primes alone at 1.09 ± 0.15 (7σ). The real character modulo 5 appears at 1.01 ± 0.07 times its prediction (14σ), with the predicted sign for each of the 24 gaps not divisible by 5. For 50 triplet patterns $(0, a, b)$, $b \le 24$, the ζ term is found at 0.98 ± 0.07, and the odd character modulo 3, which cannot appear for pairs, appears at 0.91 ± 0.13 (7σ), with opposite signs for mirror patterns such as $(0, 2, 6)$ and $(0, 4, 6)$. Characters with vanishing predicted coefficient are consistent with noise. All errors are obtained by time slides and account for the correlations between patterns. Nothing here is proved; the results are numerical evidence for a heuristic.

1. INTRODUCTION

For an admissible pattern $\mathcal{H} = (h_1, \dots, h_k)$ of distinct integers ($h_1 = 0$), the prime k-tuples conjecture of Hardy and Littlewood [5] predicts

$$(1)\quad W_{\mathcal{H}}(x) := \sum_{\substack{p \le x \\ p + h_i \text{ prime } \forall i}} \prod_{i=1}^{k} \log(p + h_i) \;\sim\; \mathfrak{S}(\mathcal{H})\, x, \qquad \mathfrak{S}(\mathcal{H}) = \prod_p \Big(1 - \frac{\nu_p(\mathcal{H})}{p}\Big)\Big(1 - \frac{1}{p}\Big)^{-k},$$

where $\nu_p(\mathcal{H})$ is the number of residue classes modulo p occupied by $\mathcal{H}$. Even the case $k = 2$, $\mathcal{H} = (0, 2)$ (twin primes) is open. This note is not about the main term, but about the *fluctuations* $W_{\mathcal{H}}(x) - \mathfrak{S}(\mathcal{H})x$, which are of size about $\sqrt{x}$ (times logarithms).

For $k = 1$, the fluctuations of the primes are described exactly by the explicit formula $\psi(x) - x = -\sum_\rho x^\rho/\rho + O(\log x)$, and in residue classes by $\psi(x, \chi) = \sum_{n \le x} \chi(n)\Lambda(n) = -\sum_{\rho_\chi} x^{\rho_\chi}/\rho_\chi + \dots$ for non-principal χ (see e.g. [3]). It is natural to expect that the fluctuations of prime patterns contain the same oscillations. For pairs, averaged over the gap h, such a formula was proposed and tested up to 10^{12} by Korevaar and te Riele [7]; related averaged formulas for the Goldbach problem go back to Fujii [4], and versions in arithmetic progressions,

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where zeros of Dirichlet L-functions appear, are studied in [10, 1]. In the opposite direction, the correlations of the zeros of ζ contain the Hardy–Littlewood constants [2, 6].

Here we look at each pattern separately and at each character separately. A one-line heuristic (Section 2) predicts the size *and the sign* of every oscillating term, without any free parameter. Two features make the test sharp:

- the coefficient of $\psi(x,\chi)$ changes sign with the pattern, so it disappears from any average over patterns and must be detected pattern by pattern;
- for pairs the coefficient vanishes for every odd character (Lemma 1), while for triplets it does not: the odd character modulo 3 must appear in triplets, with opposite signs for a pattern and its mirror image (Lemma 2).

Single patterns are noisy: for twin primes, the counting noise of W_2 is about twenty times larger than the predicted oscillation in a naive spectrogram. We use instead a matched filter, as in gravitational-wave searches: the expected waveform is known exactly (it is computed from the primes themselves, or from the zeros), and we project the data on it in a wide frequency band. Noise is estimated by time slides.

We stress that nothing below is a theorem. The prediction is heuristic, and the numerical results are evidence for it, in a limited range. We have not found the pattern-by-pattern and character-by-character prediction stated in this form in the literature, but it follows from standard heuristics and may well be known to experts.

2. The heuristic prediction

Fix a modulus q and let $A_q(\mathcal{H})$ be the set of residue classes $a \bmod q$ such that $\gcd(a+h_i,q)=1$ for all i. In the Hardy–Littlewood model, primes p completing the pattern are equidistributed among the classes of $A_q(\mathcal{H})$. Near t, the density of primes in a class r coprime to q is, by orthogonality,

$$\frac{1}{\varphi(q)}\Big(1+\sum_{\chi \bmod q}\bar\chi(r)\,\delta_\chi(t)\Big), \qquad \delta_\chi(t)=\frac{d}{dt}\big(\psi(t,\chi)-\mathbf{1}_{\chi=\chi_0}\,t\big),$$

understood as a distribution whose oscillating part is $-\sum_{\rho_\chi} t^{\rho_\chi-1}$. If each member $p+h_i$ of the pattern carries independently the fluctuation of its own class, the density of patterns is multiplied by $\prod_i(1+\sum_\chi \bar\chi(a+h_i)\delta_\chi) \approx 1+\sum_\chi\big(\sum_i \bar\chi(a+h_i)\big)\delta_\chi$. Averaging over $a\in A_q(\mathcal{H})$ and integrating gives the following.

Heuristic prediction 1. *Let $u=\log x$ and $E_{\mathcal{H}}(u)=(W_{\mathcal{H}}(x)-\mathfrak{S}(\mathcal{H})x)/\sqrt{x}$. In any fixed band of frequencies γ (in the variable u), and up to a noise term,*

(2)
$$E_{\mathcal{H}}(u) \;\approx\; \mathfrak{S}(\mathcal{H})\sum_{\chi \bmod q} c_\chi(\mathcal{H})\,\frac{\psi(x,\chi)-\mathbf{1}_{\chi=\chi_0}x}{\sqrt{x}}, \qquad c_\chi(\mathcal{H})=\frac{1}{|A_q(\mathcal{H})|}\sum_{a\in A_q(\mathcal{H})}\;\sum_{i=1}^{k}\bar\chi(a+h_i).$$

For the principal character, $c_{\chi_0}(\mathcal{H})=k$, so the zeros of ζ enter with weight $k\,\mathfrak{S}(\mathcal{H})$: this is the expansion $(1+\varepsilon)^k\approx 1+k\varepsilon$. For several moduli one may combine the characters by the Chinese remainder theorem; below we only use one character at a time, and fit all templates jointly. For a complex character, χ and $\bar\chi$ together contribute $2\mathfrak{S}\,(\operatorname{Re}c_\chi\,\operatorname{Re}\psi(x,\chi)-\operatorname{Im}c_\chi\,\operatorname{Im}\psi(x,\chi))$.

The prediction ignores the products of fluctuations (second order), the prime powers, and the correlations between the members of the pattern beyond the singular series. Its content is the size and sign of the first-order oscillating terms. The following elementary facts make it easy to test.

Lemma 1. *Let $\mathcal{H} = (0, h)$ and let q be a prime not dividing h. Then $c_\chi(\mathcal{H}) = -2\bar\chi(h)/(q-2)$ if $\chi \neq \chi_0$ is even, and $c_\chi(\mathcal{H}) = 0$ if χ is odd. If $q \mid h$, then $c_\chi(\mathcal{H}) = 0$ for $\chi \neq \chi_0$.*

Proof. If $q \nmid h$, $A_q = \{a : a \not\equiv 0, -h\}$ has $q-2$ elements. Since $\sum_{a \bmod q} \bar\chi(a) = 0$, we get $\sum_{a\in A_q} \bar\chi(a) = -\bar\chi(-h)$ and $\sum_{a\in A_q} \bar\chi(a+h) = -\bar\chi(h)$, whose sum is $-\bar\chi(h)(1+\chi(-1))$. If $q \mid h$, both sums are over all nonzero classes and vanish. □

Lemma 2. *Let $\mathcal{H}^* = h_{\max} - \mathcal{H}$ be the mirror image of $\mathcal{H}$. Then $c_\chi(\mathcal{H}^*) = \chi(-1)\, c_\chi(\mathcal{H})$. In particular $c_\chi(\mathcal{H}) = 0$ for odd χ whenever $\mathcal{H}^*$ is a translate of $\mathcal{H}$, as for every pair.*

Proof. The map $a \mapsto -a - h_{\max}$ is a bijection $A_q(\mathcal{H}) \to A_q(\mathcal{H}^*)$, and $-a - h_{\max} + (h_{\max} - h_i) = -(a + h_i)$, so each character value is multiplied by $\bar\chi(-1) = \chi(-1)$. □

Remark 1. For triplets the odd character χ_3 modulo 3 need not vanish. The pattern $(0, 2, 6)$ forces $p \equiv 2 \pmod 3$, so its members lie in the classes $2, 1, 2$ and $c_{\chi_3} = -1$; its mirror image $(0, 4, 6)$ forces $p \equiv 1$ and gives $c_{\chi_3} = +1$. On the other hand, a pattern of even integers imposes nothing modulo 4 or 8 on an odd prime, so $c_\chi = 0$ for the characters modulo 4 and 8, and for the real character modulo 12; these serve as controls.

3. Data and method

Sieve. We sieved all integers up to $N = 10^{10}$ (in Python with NumPy, and independently in C), and recorded on a grid $x_j = \lfloor e^{u_j} \rfloor$, $u_j = \log 10^3 + 0.0005\, j$, the quantities $\psi(x)$, $\psi(x, \chi)$ for the characters listed below, and $W_\mathcal{H}(x)$ for the patterns studied. Checks: $\pi(10^{10}) = 455\,052\,511$ and $27\,412\,679$ twin prime pairs below 10^{10}, the known values. A faster C sieve extends W_h for $h = 2, 4, 6$ to 10^{12}, reproducing the known count of $1\,870\,585\,220$ twin pairs below 10^{12} [9]. The last grid point below 10^{10} is $x = 9\,999\,043\,536$.

Characters used: the real characters $\chi_3, \chi_4, \chi_5, \chi_8, \chi_{12}, \chi_{13}$ (moduli $3, 4, 5, 8, 12, 13$; χ_3, χ_4 are odd, the others even), the complex odd character χ_5' modulo 5 with $\chi_5'(2) = i$, and the cubic character χ_7 modulo 7 (even).

Matched filter. Let $f(u) = (\psi(x) - x)/\sqrt{x}$ and $g_\chi(u) = \psi(x, \chi)/\sqrt{x}$ (real and imaginary parts as separate templates when χ is complex). On the interval $9 \le u \le \log x_{\max}$ we apply a Hann window, take the discrete Fourier transform, and keep the frequencies $50 \le \gamma \le 6000$ (this band contains 5 588 zeros of ζ). Below $\gamma = 50$ the counting noise of a single pattern dominates. For each pattern $\mathcal{H}$ we compute the coefficients $\hat\beta_\mathcal{H}$ of the least-squares fit of the filtered $E_\mathcal{H}$ by all filtered templates jointly, and compare them with the predictions (2). No parameter is adjusted.

Noise: time slides. The templates are shifted cyclically along u by 50 offsets between 0.75 and 13 (in units of u), which destroys the phase coherence with the true oscillations, and the fit is repeated. The standard deviation $\sigma_\mathcal{H}$ of the 50 shifted coefficients is the noise of $\hat\beta_\mathcal{H}$.

Combining patterns. For a template with predictions $\pi_\mathcal{H}$, we estimate a common ratio "measured/predicted" $\kappa = \sum_\mathcal{H} \hat\beta_\mathcal{H} \pi_\mathcal{H} \sigma_\mathcal{H}^{-2} / \sum_\mathcal{H} \pi_\mathcal{H}^2 \sigma_\mathcal{H}^{-2}$. The patterns share their primes, so their noises are correlated, and the naive error $(\sum \pi_\mathcal{H}^2 \sigma_\mathcal{H}^{-2})^{-1/2}$ is too small (by a factor of about 2 for the ζ template). We therefore recompute κ with the shifted coefficients for each of the 50 time slides and use the standard deviation of these values; we report the larger of the two errors. For templates whose prediction is 0 for every pattern, we report $\chi^2 = \sum(\hat\beta_\mathcal{H}/\sigma_\mathcal{H})^2$ and compare it with the distribution of the same statistic over the time slides.

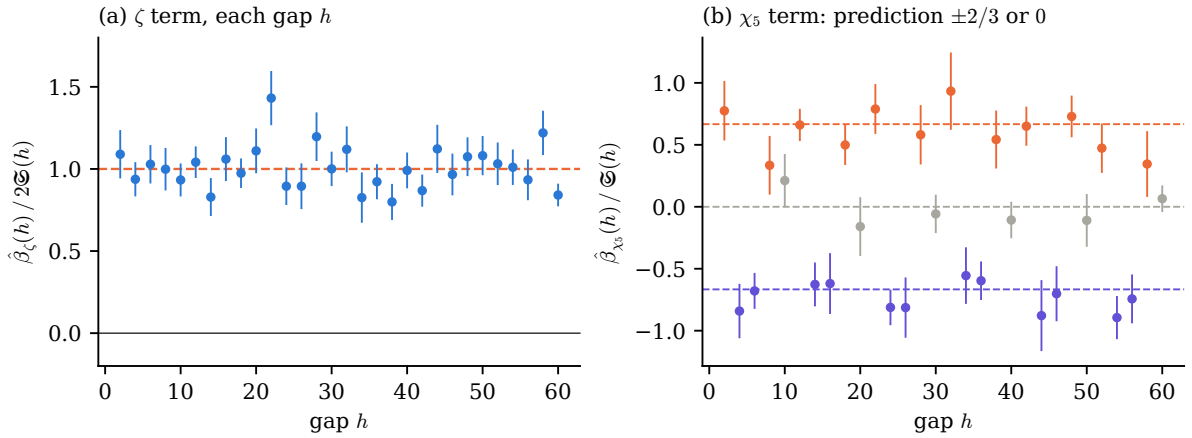


FIGURE 1. Pairs $(p, p+h)$ up to 10^{10}. (a) Coefficient of the ζ template divided by its prediction $2\mathfrak{S}(h)$. (b) Coefficient of the χ_5 template divided by $\mathfrak{S}(h)$; dashed lines: predictions $+2/3$ (orange), $-2/3$ (violet), 0 for $5 \mid h$ (grey). Error bars: one standard deviation from time slides.

4. Results for pairs

The thirty gaps $h = 2, 4, \ldots, 60$ all follow (1) to within 0.016% at 10^{10}. Table 1 and Figure 1 give the coefficients.

- *The ζ term.* Combining the thirty gaps, $\kappa_\zeta = 0.98 \pm 0.04$ (22σ). Each gap separately is consistent with 1; gaps with $6 \mid h$, whose $\mathfrak{S}(h)$ is twice as large, carry an oscillation twice as large.
- *Twin primes alone.* For $h = 2$, $\hat{\beta}/2\mathfrak{S}(2) = 1.09 \pm 0.15$ (7σ); none of the 50 time slides exceeds 0.37.
- *Characters.* χ_5: $\kappa = 1.01 \pm 0.07$ (14σ), with the predicted sign ($+2/3$ for $h \equiv \pm 2$, $-2/3$ for $h \equiv \pm 1 \bmod 5$) for each of the 24 gaps not divisible by 5, and values consistent with 0 for the six multiples of 5. Cubic χ_7 (real and imaginary parts together): 1.08 ± 0.08 (14σ). χ_{13}, whose coefficients are small ($\pm 2/11$): 0.93 ± 0.17 (5σ).
- *Controls.* The odd characters χ_3, χ_4, χ_5' and the even characters χ_8, χ_{12}, for which the prediction vanishes, give χ^2 between 26 and 37 for 30 gaps, while the time slides give 31 ± 9.

A blind test beyond 10^{10}. The prediction being fixed, we counted the new pairs with $10^{10} < p \le 10^{12}$ for $h = 2, 4, 6$. Since $\psi(x)$ was not computed that far, the template was built from the zeros, $f(u) = -2\,\mathrm{Re}\sum_{\gamma<7000} e^{i\gamma u}/(\frac{1}{2} + i\gamma)$, using 138 069 zeros computed with the Riemann–Siegel formula; on $[10^3, 10^{10}]$ its correlation with the sieve template in the band is 0.991. On the new data alone: 0.61 ± 0.31 $(h = 2)$, 1.07 ± 0.31 $(h = 4)$, 1.05 ± 0.29 $(h = 6)$, and 0.91 ± 0.17 combined (5σ). The signal-to-noise ratio does not grow much with x: the filtered counting noise of $W_2/\sqrt{x}$ grows roughly like $\log x$ (from 0.77 to 1.35 between 10^5 and 10^8), while the oscillation keeps a constant size. For twin primes alone, the significance reaches about 8σ near $x = 5 \cdot 10^8$ and is 6–7σ from 10^{10} to $5 \cdot 10^{11}$.

TABLE 1. Pairs up to 10^{10}: combined ratio κ (measured/predicted) for each template. For controls (prediction 0 for all gaps), χ^2 for 30 gaps and its mean $\pm$ standard deviation over the time slides.

template	prediction	result
ζ	$2\mathfrak{S}(h)$	$\kappa = 0.980 \pm 0.044$
χ_5 (even)	$\mp\frac{2}{3}\mathfrak{S}(h)$ or 0	$\kappa = 1.006 \pm 0.070$
χ_7 cubic (even), Re and Im	$-\frac{2}{5}\mathfrak{S}(h)\bar{\chi}_7(h)$	$\kappa = 1.083 \pm 0.079$
χ_{13} (even)	$\mp\frac{2}{11}\mathfrak{S}(h)$ or 0	$\kappa = 0.925 \pm 0.172$
χ_3 (odd)	0	$\chi^2 = 28.0$ (slides 30.5 ± 7.8)
χ_4 (odd)	0	$\chi^2 = 37.1$ (slides 31.1 ± 7.3)
χ_5' (odd), Re / Im	0	$\chi^2 = 26.3$ / 30.7 (slides 31.1 ± 7.9 / 30.6 ± 10.1)
χ_8 (even)	0	$\chi^2 = 25.9$ (slides 31.2 ± 10.2)
χ_{12} (even)	0	$\chi^2 = 34.0$ (slides 30.8 ± 10.5)

5. Results for triplets

We studied the 50 admissible triplet patterns $(0, a, b)$ with $a < b \le 24$ even (admissibility only matters modulo 2 and 3), and the quadruplet $(0, 2, 6, 8)$; in total about 190 million triplets up to 10^{10}.

- *Hardy–Littlewood.* All 50 counts agree with $\mathfrak{S}(\mathcal{H}) \int_2^x dt/\prod_i \log(t+h_i)$ to within 0.11%. For instance there are 2 713 122 triplets $(p, p+2, p+6)$ with $p \le 9\,999\,043\,536$, against 2 715 062 predicted ($\mathfrak{S} = 2.8582$).
- *The ζ term* (prediction $3\mathfrak{S}(\mathcal{H})$): $\kappa = 0.98 \pm 0.07$ (14σ).
- *The odd character χ_3.* The prediction is $-\mathfrak{S}$ for 22 patterns, $+\mathfrak{S}$ for 22 others, and 0 for the 6 patterns made of multiples of 6. We find $\kappa = 0.91 \pm 0.13$ (7σ). Individual patterns are noisy: $(0, 2, 6)$ gives -0.70 ± 0.88 and its mirror image $(0, 4, 6)$ gives $+2.20 \pm 1.07$ (in units of $\mathfrak{S}$).
- *Other characters.* χ_5: 1.01 ± 0.10 (10σ); cubic χ_7: 0.92 ± 0.13 (7σ); the odd complex character χ_5', whose prediction is nonzero for 20 of the 50 patterns: 0.92 ± 0.29 (3σ); χ_{13}: 0.64 ± 0.39, inconclusive.
- *Controls.* $\chi_4, \chi_8, \chi_{12}$ (prediction 0 for every triplet): $\chi^2 = 45.4$, 51.4, 53.9 for 50 patterns, against 51.5 ± 11.0, 52.1 ± 9.8, 51.8 ± 13.4 over the time slides.
- *The quadruplet* $(0, 2, 6, 8)$: 180 517 up to the same bound, 0.3% below the prediction; its ζ coefficient, 1.9 ± 0.8 times $4\mathfrak{S}$, is too noisy to be informative.

The naive error for the ζ template, which treats the 50 patterns as independent, would be ± 0.034 (29σ); the time-slide error, which accounts for their correlation, is twice as large.

6. Discussion and limitations

(1) The prediction (2) is a first-order heuristic. A derivation along the lines of the circle method, with characters, would be welcome, as would a treatment of the second-order terms. Nothing here bears on the existence of infinitely many twin primes or prime triplets.

(2) The test is restricted to $x \le 10^{10}$ (and 10^{12} for three gaps), to small moduli, and to one frequency band. Because the counting noise grows like $\log x$ after normalisation by $\sqrt{x}$, larger x helps little; more patterns and more characters help more.

(3) The time-slide noise estimate assumes approximate stationarity of the noise along u; with 50 slides, the errors themselves are known to about 10%.

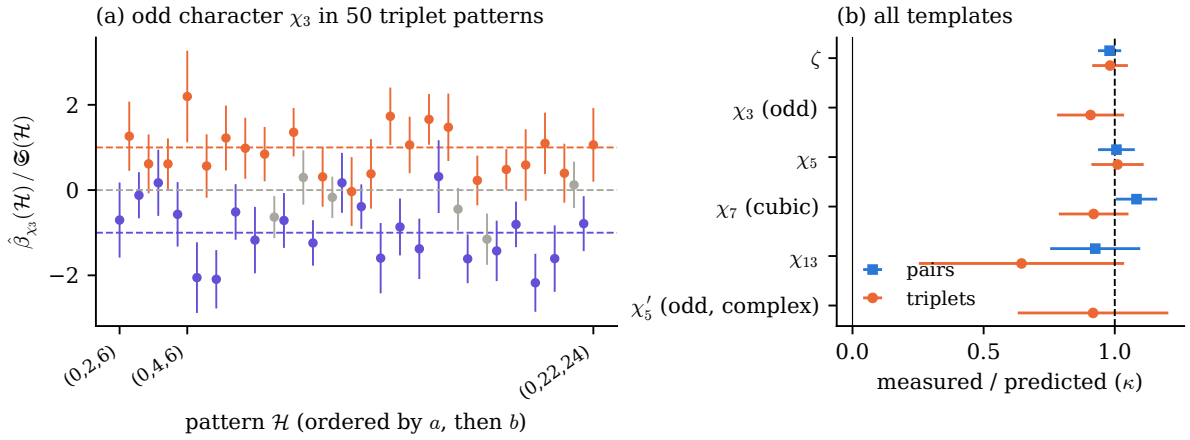


FIGURE 2. (a) Triplet patterns $(0, a, b)$, $b \le 24$, up to 10^{10}: coefficient of the χ_3 template divided by $\mathfrak{S}(\mathcal{H})$. Predictions: -1 (violet), $+1$ (orange), 0 (grey). (b) Combined ratio measured/predicted for every template with a nonzero prediction, for pairs (squares) and triplets (circles); χ_3 and χ_5' have zero prediction for pairs.

TABLE 2. Selected triplet patterns up to $x = 9\,999\,043\,536$: counts, Hardy–Littlewood prediction, singular series, and the coefficients of the ζ template (divided by $3\mathfrak{S}$) and of the χ_3 template (divided by $\mathfrak{S}$), with the prediction c_{χ_3}.

$\mathcal{H}$	count	HL prediction	$\mathfrak{S}(\mathcal{H})$	ζ	c_{χ_3}	χ_3
$(0, 2, 6)$	2 713 122	2 715 062	2.8582	0.93 ± 0.28	-1	-0.70 ± 0.88
$(0, 4, 6)$	2 712 007	2 715 062	2.8582	1.86 ± 0.34	$+1$	$+2.20 \pm 1.07$
$(0, 2, 8)$	2 715 324	2 715 062	2.8582	0.79 ± 0.33	$+1$	$+1.26 \pm 0.81$
$(0, 6, 8)$	2 715 436	2 715 061	2.8582	0.73 ± 0.34	-1	-1.17 ± 0.78
$(0, 6, 12)$	5 427 503	5 430 121	5.7165	0.96 ± 0.22	0	-0.63 ± 0.49
$(0, 12, 24)$	5 428 556	5 430 119	5.7165	0.89 ± 0.20	0	-1.15 ± 0.60
$(0, 22, 24)$	3 055 865	3 054 441	3.2155	1.51 ± 0.34	$+1$	$+1.06 \pm 0.86$

(4) The templates $\psi(x, \chi)$ include prime powers, while $W_{\mathcal{H}}$ does not; the corresponding smooth terms (of size $\sqrt{x}$ and below) lie outside the band and do not affect the fit.
(5) The averaged ζ term for pairs is the formula of Korevaar and te Riele [7]. The character terms cancel in such averages since their signs change with the pattern, which is why they must be detected pattern by pattern. Analogous character sums appear in averaged Goldbach problems in arithmetic progressions [10, 1]. The variance of primes in short intervals is related to the same singular series [8].

DATA AND CODE AVAILABILITY

The sieves (Python and C), the zero computation, the analysis scripts and the list of patterns are available, under the MIT license, in the folder `exploration/calculs` of the repository

https://github.com/GEREMIAA/Riemann_Other_Ways

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References

[1] G. Bhowmik, K. Halupczok, K. Matsumoto, Y. Suzuki, *Goldbach representations in arithmetic progressions and zeros of Dirichlet L-functions*, preprint, arXiv:1704.06103.

[2] E. B. Bogomolny, J. P. Keating, *Gutzwiller's trace formula and spectral statistics: beyond the diagonal approximation*, Phys. Rev. Lett. **77** (1996), 1472–1475.

[3] H. Davenport, *Multiplicative Number Theory*, 3rd ed., Graduate Texts in Mathematics 74, Springer, 2000.

[4] A. Fujii, *An additive problem of prime numbers*, Acta Arith. **58** (1991), 173–179.

[5] G. H. Hardy, J. E. Littlewood, *Some problems of 'Partitio numerorum'; III: On the expression of a number as a sum of primes*, Acta Math. **44** (1923), 1–70.

[6] J. P. Keating, D. J. Smith, *Twin prime correlations from the pair correlation of Riemann zeros*, preprint (2019), arXiv:1903.07057.

[7] J. Korevaar, H. te Riele, *Average prime-pair counting formula*, Math. Comp. **79** (2010), 1209–1229, arXiv:0902.4352.

[8] H. L. Montgomery, K. Soundararajan, *Primes in short intervals*, Comm. Math. Phys. **252** (2004), 589–617, arXiv:math/0409258.

[9] OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, sequence A007508 (number of twin prime pairs below 10^n), `https://oeis.org/A007508`.

[10] Y. Suzuki, *A mean value of the representation function for the sum of two primes in arithmetic progressions*, preprint, arXiv:1504.01967.

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